Uniqueness and Operator Realization on Hyperbolic Logarithmic Lattices

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Abstract

Let Γ be a hyperbolic Coxeter group acting properly discontinuously and cocompactly on $\mathbb{H}^3$. We assign to each group element γ a logarithmic weight $q(\gamma) = \kappa\, d(0, \gamma \cdot 0)$. We prove three structural results: (1) each boundary direction $\xi \in \partial\mathbb{H}^3$ admits a unique minimal representative in Γ (torsion-free case: exactly unique); (2) there is a uniform positive gap $\Delta > 0$ between any two distinct values in the weight spectrum; (3) integer coordinates defined via the Milnor–Švarc quasi-isometry are stable under bounded deformations of the boundary configuration. All results follow from hyperbolicity, proper discontinuity, and the classification of isometries of $\mathbb{H}^3$.

1 Introduction

Discrete logarithmic hierarchies appear in warped geometries, renormalization group flows, and bulk-boundary correspondences. A physically useful framework must explain why each weight supports a unique dominant contribution, why subleading contributions are suppressed by a definite amount, and why the organization is stable under perturbations. We show all three properties follow from the geometry of a cocompact group action on hyperbolic space.

Our three main results are: **uniqueness** (Theorem 4.1), **uniform gap** (Lemma 5.1), and **coordinate rigidity** (Theorem 7.2).

2 Geometric Setting

Assumption 2.1. Γ is torsion-free, acting properly discontinuously and cocompactly on $\mathbb{H}^3$ by orientation-preserving isometries. Fix basepoint $0 \in \mathbb{H}^3$.

For $\gamma \in \Gamma$ set $q(\gamma) = \kappa\, d(0, \gamma \cdot 0)$, $\kappa > 0$.

Definition 2.2 (Isometry types)**.** An isometry $\delta \in \mathrm{Isom}(\mathbb{H}^3)$ is *elliptic* if it fixes a point in $\mathbb{H}^3$; *parabolic* if it fixes exactly one point on $\partial\mathbb{H}^3$ and no point in $\mathbb{H}^3$; *hyperbolic* if it fixes exactly two points on $\partial\mathbb{H}^3$ and translates along the geodesic connecting them.

Lemma 2.3 (Parabolic elements increase distance)**.** *A cocompact group Γ contains no parabolic elements [2].*

3 Boundary Directions and Minimal Representatives

Definition 3.1 (Boundary direction)**.** A *boundary direction* is a point $\xi \in \partial\mathbb{H}^3 \cong \mathbb{S}^2$. We say $\gamma \in \Gamma$ *converges to* ξ if the geodesic ray from 0 through $\gamma \cdot 0$ has ξ as its ideal endpoint.

Definition 3.2 (Minimal representative)**.** $\gamma \in \Gamma$ is a *minimal representative* for ξ if it converges to ξ and $q(\gamma) = \inf\{q(\gamma') : \gamma' \text{ converges to } \xi\}$.

4 Uniqueness of Minimal Representatives

Theorem 4.1 (Uniqueness)**.** *Let $\gamma_1, \gamma_2 \in \Gamma$ be two minimal representatives for $\xi \in \partial\mathbb{H}^3$. Then $\delta := \gamma_1^{-1}\gamma_2$ is elliptic. Since Γ is torsion-free, $\delta = \mathrm{id}$ and $\gamma_1 = \gamma_2$.*

Proof. Since γ_1 and γ_2 both converge to ξ, the element $\delta = \gamma_1^{-1}\gamma_2$ fixes $\xi \in \partial\mathbb{H}^3$. An isometry fixing a boundary point is either elliptic or parabolic. Since Γ is cocompact it contains no parabolic elements (Lemma 2.3). Hence δ is elliptic. Since Γ is torsion-free the only elliptic element is the identity, so $\delta = \mathrm{id}$ and $\gamma_1 = \gamma_2$. □

5 The Uniform Suppression Gap

Lemma 5.1 (Uniform gap)**.** *There exists $\Delta > 0$ such that any two distinct elements $\gamma, \gamma' \in \Gamma$ satisfy $|q(\gamma) - q(\gamma')| \geq \Delta$.*

Proof. The orbit $\Gamma\cdot 0$ is discrete by proper discontinuity. The systole $\Delta_0 = \min_{\gamma \neq \mathrm{id}} d(0, \gamma\cdot 0) > 0$, and $\Delta = \kappa\Delta_0$. □

Corollary 5.2 (Uniform suppression)**.** *Every subleading operator satisfies $q(\gamma') \geq q(\gamma) + \Delta$.*

6 Integer Coordinates and the Milnor–Švarc Quasi-isometry

Definition 6.1 (Integer coordinates)**.** Fix a finite generating set $\mathcal{S} = \{s_1, s_2, s_3\}$ for Γ. The integer coordinates $(a, b, c) \in \mathbb{Z}^3$ of γ are the net counts of each generator in a reduced word: $a = n_1(\gamma) - n_1^-(\gamma)$, etc.

Proposition 6.2 (Milnor–Švarc quasi-isometry)**.** *Since Γ acts properly discontinuously and cocompactly on $\mathbb{H}^3$, the orbit map $\gamma \mapsto \gamma \cdot 0$ is a quasi-isometry $(\Gamma, d_{\mathcal{S}}) \to (\mathbb{H}^3, d)$ [1].*

7 Coordinate Rigidity under Bounded Deformations

Definition 7.1 (Bounded deformation)**.** A *bounded deformation of magnitude ε* perturbs each boundary direction $\xi(f)$ to $\xi'(f)$ with $d_{\mathbb{S}^2}(\xi(f), \xi'(f)) \leq \varepsilon$.

Theorem 7.2 (Coordinate rigidity)**.** *For $\varepsilon \leq \varepsilon_0 := \frac{1}{2}\min_{f \neq f'} d_{\mathbb{S}^2}(\xi(f), \xi(f'))$, each minimal representative γ_ξ and its integer coordinates (a, b, c) are unchanged.*

Proof. For $\varepsilon \leq \varepsilon_0$, perturbed directions $\xi'(f)$ and $\xi'(f')$ remain distinct and each $\xi'(f)$ lies closer to $\xi(f)$ than to any other direction. Hence the minimal representative for $\xi'(f)$ equals that for $\xi(f)$. $\square$

8 Conclusion

Hyperbolic logarithmic lattices arising from cocompact group actions on $\mathbb{H}^3$ are rigid in three ways: unique minimal representatives, uniform positive gap between weight values, and stable integer coordinates under small deformations. These properties hold for any cocompact torsion-free group and require no external dynamical assumptions.

References

[1] M. R. Bridson and A. Haefliger. *Metric Spaces of Non-Positive Curvature*. Springer, 1999.

[2] J. G. Ratcliffe. *Foundations of Hyperbolic Manifolds*. Springer, 2006.